Numbers, Percents & Ratios

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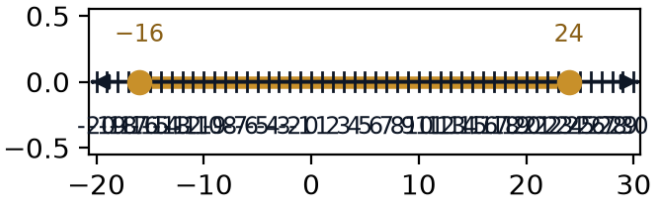
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the complex number system, where $i^2 = -1$, the expression $\frac{a+6i}{2-i}$ is equal to $4 + bi$ for some real numbers a and b. What is the value of $a + b$?

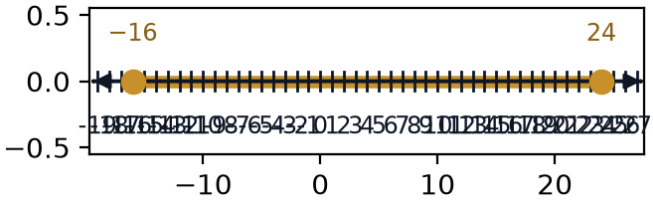
2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

On the real number line below, point A has coordinate -16 and point C has coordinate 24. Point B has coordinate k, where $-16 < k < 24$. If the ratio of the distance between A and B to the distance between B and C is $3 : 5$, what is the value of k?



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

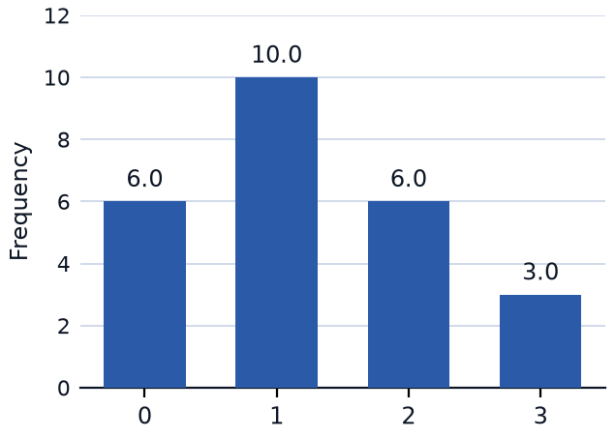
On the number line shown below, point A is located at -16 and point B is located at 24. Point P lies on the line segment connecting A and B such that the ratio of the length of $\overline{AP}$ to the length of $\overline{PB}$ is $3 : 2$. What is the coordinate of point P?



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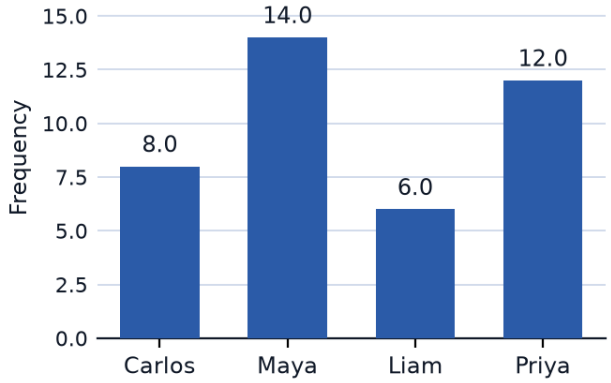
4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The bar graph below shows the distribution of the number of siblings reported by the 25 students in a homeroom class. If 1 student is selected at random from the class, what is the probability that the selected student has at least 2 siblings?



5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

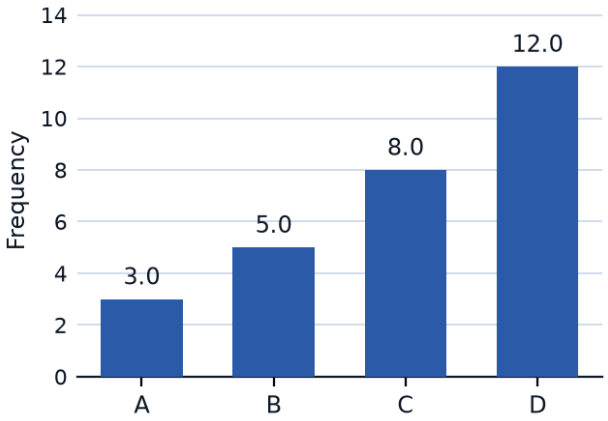
The bar graph below shows the number of hours four students volunteered at a community center last week. What percent of the total volunteer hours logged by the four students was logged by Maya?



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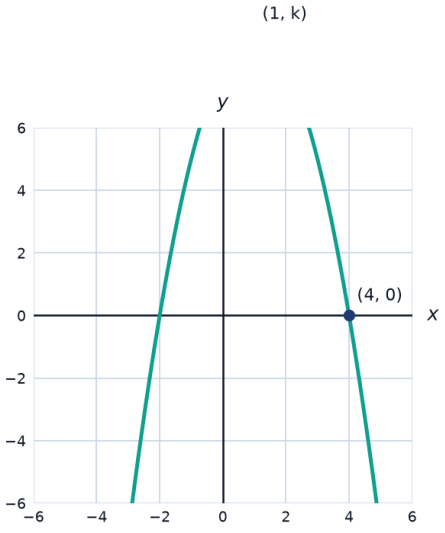
6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The bar graph below shows the number of books read by 4 students—Student A, Student B, Student C, and Student D—during summer break. Student A read 3 books, Student B read 5 books, Student C read 8 books, and Student D read 12 books. What is the average (arithmetic mean) number of books read by the 4 students?



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the quadratic function $f(x) = -x^2 + bx + c$ is shown in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(1, k)$, and the graph passes through the point $(4, 0)$. What is the value of k?

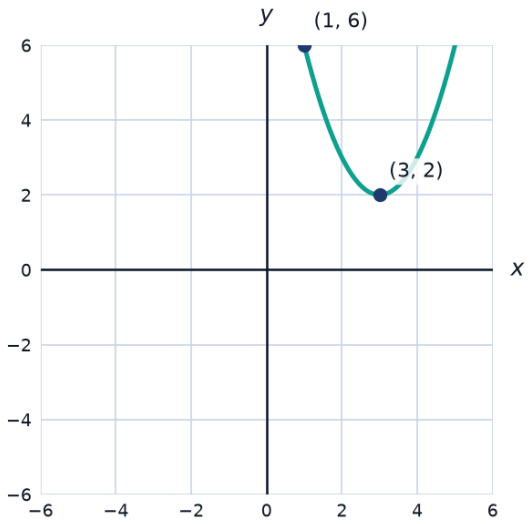


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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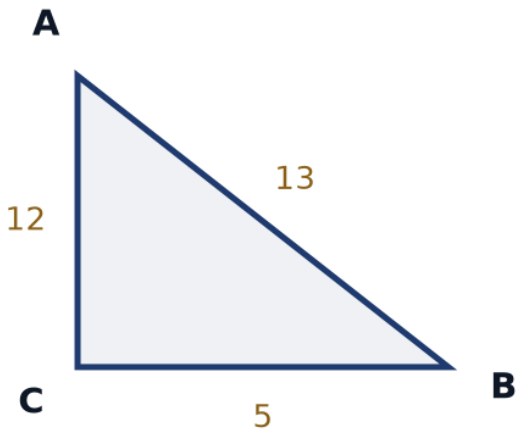
8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown in the standard (x, y) coordinate plane below, with its vertex at $(3, 2)$ and passing through the point $(1, 6)$. What is the value of b?



9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

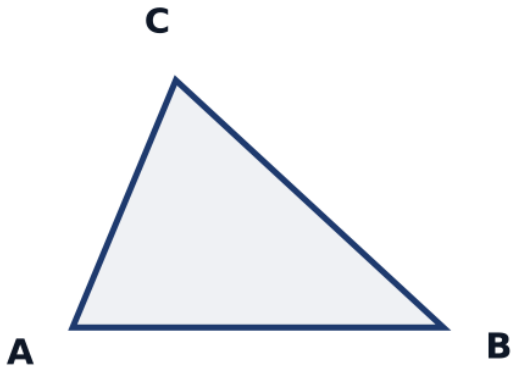
In the right triangle ΔABC shown below, the right angle is at vertex C. The lengths of the sides are given in inches: $AC = 12$, $BC = 5$, and $AB = 13$. What is the value of $\cos B$?



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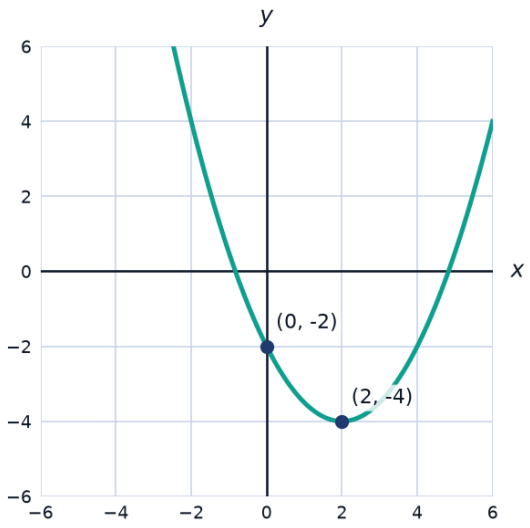
10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC shown below, the measure of $\angle A$ is 65° and the measure of $\angle B$ is 45°. What is the value of x ?



11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

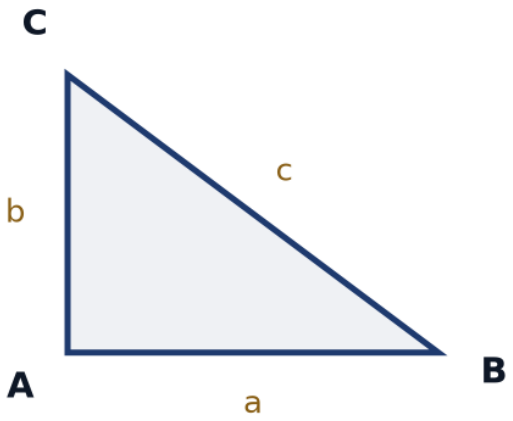
The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -4)$ and y-intercept $(0, -2)$. If the point $(6, c)$ lies on the graph, what is the value of c?



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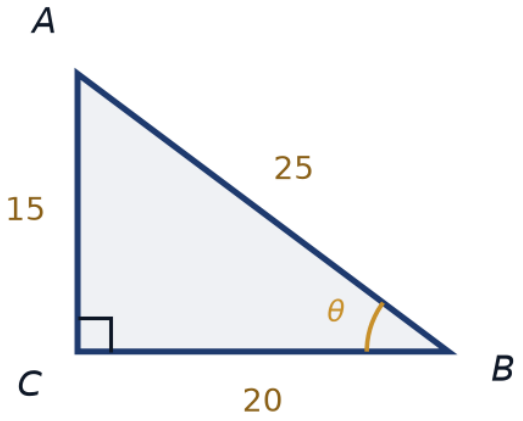
12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has legs 9 and 12. Find the hypotenuse.



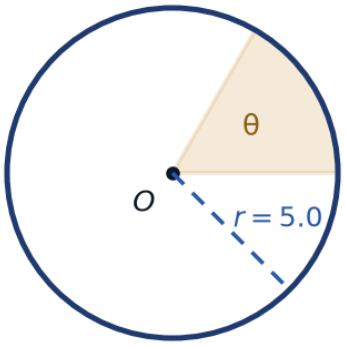
13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A triangle has angles 50° and 60°. Find the third.



14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the area of a circle with radius 5.



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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

The quadratic function $f(x) = x^2 - 2kx + (3k + 10)$ has 2 distinct real zeros. If both zeros are strictly greater than 2, how many integer values of k are possible?

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Functions & Graphs

The graph of the quadratic function $f(x) = 2x^2 - 12x + c$ in the standard (x, y) coordinate plane passes through the point $(1, -3)$. What is the minimum value of $f(x)$?

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

In the figure below, ΔABC is a right triangle with right angle at C, and altitude $\overline{CD}$ of length h is perpendicular to hypotenuse $\overline{AB}$. If $AD = 4$ and $DB = 9$, what is the area of ΔABC?

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Plane Geometry

If $\cos\theta = \frac{12}{13}$ and $0^\circ < \theta < 90^\circ$, what is the value of $\tan\theta$?

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

In the figure below, ΔABC is a right triangle with the right angle at B, and $\overline{BD}$ is an altitude to hypotenuse $\overline{AC}$. The length of $\overline{AD}$ is 4 units and the length of $\overline{DC}$ is 9 units. What is the area, in square units, of ΔABC?

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Statistics & Probability

In right triangle ΔABC, the measure of $\angle C$ is 90°. The length of side AC is 12 and the length of side BC is 5. What is the value of $\cos A$?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the value of the numerical expression $|-14| - 3|5-9| + (-2)^3$?

a. -6

b. -10

c. 2

d. 14

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If p is a positive prime number and q is a negative integer, which of the following values must always be negative?

a. $p - q$

b. $q - p$

c. $p^2 + q^2$

d. $\frac{p}{q^2}$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the greatest common factor of 72, 108, and 180?

a. 18

b. 24

c. 36

d. 72

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

On a blueprint, a scale of $\frac{1}{4}$ inch represents 3 feet. If a conference room has a length of 36 feet, what is its length on the blueprint?

a. 2.5 inches

b. 3 inches

c. 4.5 inches

d. 6 inches

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A car travels 180 miles on 6 gallons of fuel. At the same rate of fuel consumption, how many gallons of fuel are required to travel 450 miles?

a. 12

b. 15

c. 16.5

d. 18

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The ratio of copper to zinc in a brass alloy is $5:3$ by weight. If a piece of brass weighs 56 kilograms, how many kilograms of copper does it contain?

a. 21

b. 30

c. 35

d. 40

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A store purchases a bicycle for 150 and marks up the price by 40%. During a weekend clearance sale, the bicycle is discounted by 20%. What is the final sale price?

a. 168

b. 174

c. 180

d. 192

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If 35% of a number N is equal to 84, what is 60% of N?

a. 120

b. 144

c. 168

d. 240

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The price of an airline ticket increased from 250 to 320. What was the percent increase in the ticket price?

a. 22%

b. 25%

c. 28%

d. 30%

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the value of $(3.2 \times 10^{7}) \times (2.5 \times 10^{-3})$ expressed in standard scientific notation?

a. 8.0×10^{4}

b. 8.0×10^{5}

c. 8.0×10^{3}

d. 0.8×10^{5}

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which of the following is equivalent to $\frac{x^5 \cdot x^{-2}}{(x^3)^2}$ for all positive values of x?

a. x^{-3}

b. x^{-1}

c. x^{0}

d. x^{2}

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Simplify the radical expression $\sqrt{75} - 2\sqrt{27} + \sqrt{48}$.

a. $2\sqrt{3}$

b. $3\sqrt{3}$

c. $4\sqrt{3}$

d. $5\sqrt{3}$

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Every student in the graduating class of Oakridge High School plans to do exactly one of the following after graduation: attend a 2-year college, attend a 4-year university, or enter the workforce directly. • The ratio of the number of students planning to attend a 2-year college to the number of students planning to attend a 4-year university is $3:5$. • Of the students planning to attend a 2-year college, exactly 80% will attend an in-state college. • Of the students planning to attend a 4-year university, exactly 72% will attend an in-state university. • The total number of students planning to attend an in-state college or university accounts for exactly 45% of the entire graduating class. If the number of students in every category and subcategory must be a positive integer, what is the minimum possible number of students in the graduating class?

a. 40

b. 100

c. 120

d. 200

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which of the following is a prime number?

a. 43

b. 36

c. 38

d. 44

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35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A boutique owner purchases a designer jacket at a wholesale price of W dollars and marks up the price by $p\%$. During an end-of-season clearance sale, this marked-up price is discounted by $p\%$. The final clearance price is $x84lessthantheoriginalwholesaleprice, andtheclearancepriceis\336$ less than the marked-up price. What is the value of W, the original wholesale price?

a. \672

b. \756

c. \1,008

d. \1,344

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a high school choir of 60 students, 40% are sopranos, 25% are altos, and the remaining students are tenors or basses. How many students in the choir are tenors or basses?

a. 15

b. 21

c. 24

d. 39

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A high school track team has 45 members, and the ratio of sprinters to distance runners is 4 to 5. If every member is either a sprinter or a distance runner, how many members of the team are distance runners?

a. 20

b. 25

c. 30

d. 36

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Plane Geometry

How many positive factors does 72 have?

a. 15

b. 17

c. 7

d. 12

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Trigonometry

After a 25\% markup, an item costs $x25$. What was the original price?

a. $x118$

b. $x182$

c. $x196$

d. $x20$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Statistics & Probability

After a 10\% discount, an item costs $x63$. What was the original price?

a. $x151$

b. $x70$

c. $x22$

d. $x86$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

A total of 70 is divided in the ratio 2:5:7. Find the middle share.

a. 47

b. 65

c. 4

d. 25

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

A total of 48 is divided in the ratio 1:3:4. Find the smallest share.

a. 29

b. 40

c. 6

d. 36

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Functions & Graphs

The sum of 9 consecutive integers is 342. What is the smallest of these integers?

a. 34

b. 23

c. 30

d. 43

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

Split 12 in the ratio 1:3. Find the smaller part.

a. 11

b. 5

c. 7

d. 3

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Plane Geometry

Split 15 in the ratio 1:2. Find the larger part.

a. 10

b. 2

c. 12

d. 4

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